KisanSaathi — Hardware Prototype Build Guide

Compact solar-powered soil sensor device — step-by-step implementation plan for the hardware team

Project context: This device is the field-sensing unit for the KisanSaathi platform. It measures soil N-P-K, pH, EC, moisture, and temperature, plus ambient air conditions, and uploads readings to the KisanSaathi backend. Data feeds into the Crop Recommendation screen, replacing the current hardcoded ICAR district averages with live, team-measured values.

Target Form Factor

- Housing dimensions: **100 × 70 × 35 mm** (size of a phone)
- Probe length: **150 mm** stainless steel (embedded into soil)
- Total length with probe: **~250 mm**
- Weight: **~200 g**
- Solar panel: **integrated flush into lid** — no external wiring
- Ingress protection: **IP67**
- Battery life (no sun): **9+ days**

```
┌──────────────────────┐   ← Solar panel (80×60mm) top-mounted
│ ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ │     epoxy-sealed into lid
├──────────────────────┤
│                      │   ← Electronics housing
│   100×70×35mm        │       Contains: PCB + battery + MPPT
│   IP67 enclosure     │
└──────────────────────┘
          │              ← Short cable (waterproof gland)
          │              ← Probe head (plastic)
          ┌──────┐
          │      │         ← Stainless steel probe (150mm)
          │      │           Measures: N, P, K, pH, EC, moisture, temperature
          │      │
          └──────┘
```

Power Budget (Why It Works)

Solar input (60×60 mm 1W panel, India avg): 5 peak sun hours × 1W × 70% efficiency = ~3.5 Wh/day → ~950 mAh/day into the 3.7V battery.

State	Current	Duration	Energy per cycle
ESP32 deep sleep	10 µA	14m 50s	negligible
Sensor read	70 mA	3 s	~2 mWh
WiFi + HTTPS POST	160 mA	5 s	~7 mWh
Daily total (96 cycles)			**~230 mAh/day**

Net: +720 mAh/day surplus. A 2000 mAh battery gives 9+ days of dark-weather runtime with healthy margin.

STEP 1 — Procure Components (Day 1–3)

Order from **Robu.in**, **ElectronicsComp**, or **Amazon India**. Keep quantities 2× for spares.

Core Shopping List (Prototype v1 — breadboard friendly)

#	Item	Qty	Where	₹ (approx)
1	ESP32-WROOM-32 DevKit V1 (38-pin)	2	Robu.in	500 each
2	JXBS-3001-NPK-TR-ECTH 7-in-1 soil probe (RS485)	1	Robu.in / Moisture Meter India	6,500
3	MAX485 TTL-RS485 module	2	Robu.in	50 each
4	USB to RS485 adapter (FTDI or CH340)	1	Robu.in (PC-side testing)	300
5	BME280 breakout (I2C)	1	Robu.in	280
6	6V 2W Monocrystalline solar panel (110×60 mm)	1	Robu.in	350
7	CN3791 MPPT solar LiPo charger module	1	Robu.in	120
8	TP4056 charging module (backup option)	1	Robu.in	30
9	3.7V 2000mAh LiPo battery w/ JST-PH connector	1	Robu.in	450
10	MT3608 boost / MP1584 buck converter module	1	Robu.in	80
11	Breadboard 830-point	1	any	150
12	Male-male + male-female jumper wire kit	1	any	200
13	USB-C cable + 5V 2A adapter	1	any	250

Total: ~₹9,300

Tools Required Before Starting

- Soldering iron (25–40W) + lead-free solder + flux
- Multimeter (basic Fluke-clone is fine)
- Wire strippers, tweezers, side cutters
- Laptop with USB-C port
- Heat shrink tubing + hot glue gun
- Shoebox of soil for bench testing (with known moisture level)

STEP 2 — Verify Each Component Individually (Day 4–5)

> Do not skip this step. Test every part works before connecting anything together. Finding a dead component later in integration costs hours of debugging.

2a. ESP32 health check

1. Install Arduino IDE 2.x
2. Add ESP32 board package: `https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json`
3. Select board: "ESP32 Dev Module"
4. Upload Blink example — onboard LED should blink
5. Open Serial Monitor at 115200 — see boot messages

✅ **Pass → ESP32 works**

2b. Soil probe health check (PC-side, using USB-RS485 adapter)

1. Connect probe → USB-RS485 adapter → laptop
2. Download "Modbus Poll" (free trial) or use Python `minimalmodbus`
3. Probe default settings: **9600 baud, 8N1, slave ID 0x01**
4. Read registers:
   - `0x0000` → soil moisture (÷10 for %)
   - `0x0001` → soil temperature (÷10 for °C)
   - `0x0002` → EC (µS/cm)
   - `0x0003` → pH (÷10)
   - `0x0004` → nitrogen (mg/kg)
   - `0x0005` → phosphorus (mg/kg)
   - `0x0006` → potassium (mg/kg)
5. Stick probe in dry pot → readings should be low moisture. Water it → moisture rises in 30 sec.

✅ **Pass → Probe works, register map confirmed**

2c. Solar panel check

1. Point at sun (midday)
2. Multimeter across output → should read **6.0–7.2V open circuit**
3. Short-circuit current (brief): ~300–350 mA

✅ **Pass → Solar panel works**

2d. Battery + charger check

1. Connect battery to CN3791 BAT+/BAT– terminals
2. Connect solar panel to SOLAR+/SOLAR– terminals
3. In sun → CHRG LED lights up

4. Multimeter on battery → voltage slowly rises toward 4.2V

✅ **Pass → Charging circuit works**

2e. BME280 check

1. Wire to ESP32: VCC→3.3V, GND→GND, SDA→GPIO21, SCL→GPIO22
2. Install `Adafruit BME280` library
3. Run example sketch → serial should print temp/humidity/pressure

✅ **Pass → Air sensor works**

STEP 3 — Breadboard Integration (Day 6–7)

Wire everything on a breadboard. **Power via USB for now — no battery yet.**

```
                    ESP32 DevKit
                  ┌──────────────────┐
Solar 6V ─────────┤ (VIN via MPPT)   │
                  │                  │
BME280 SDA ───────┤ GPIO21 (SDA)     │
BME280 SCL ───────┤ GPIO22 (SCL)     │
BME280 VCC ───────┤ 3.3V             │
BME280 GND ───────┤ GND              │
                  │                  │
                  │ GPIO17 (TX2) ────┼── MAX485 DI
                  │ GPIO16 (RX2) ────┼── MAX485 RO
                  │ GPIO4 (DE+RE) ───┼── MAX485 DE & RE tied
                  │ 5V ──────────────┼── MAX485 VCC
                  │ GND ─────────────┼── MAX485 GND
                  │                  │
                  │                  │   MAX485
                  │                  │
                  │                  │   ┌─────┐
                  │                  │   │  A  ├── Probe RS485 A (yellow)
                  │                  │   │  B  ├── Probe RS485 B (blue)
                  │                  │   └─────┘
                  │                  │
                  │                  │   Probe power:
                  │                  │   +12V ← boost from battery
                  │                  │   GND
                  └──────────────────┘
```

Critical: The 7-in-1 probe needs **12V power** (not 3.3V or 5V). Use MT3608 boost converter: input 3.7V from battery, output set to 12V via trimpot. Measure with multimeter before connecting probe.

STEP 4 — Firmware Bring-Up (Day 8–10)

Follow a layered approach. Get each layer working before moving on.

4a. Read soil probe from ESP32

```
#include <ModbusMaster.h>
ModbusMaster node;
void preTransmission()  { digitalWrite(4, 1); }
void postTransmission() { digitalWrite(4, 0); }

void setup() {
  Serial.begin(115200);
  Serial2.begin(9600, SERIAL_8N1, 16, 17);
  pinMode(4, OUTPUT);
  node.begin(1, Serial2);
  node.preTransmission(preTransmission);
  node.postTransmission(postTransmission);
}

void loop() {
  if (node.readHoldingRegisters(0x0000, 7) == node.ku8MBSuccess) {
    Serial.printf("Moisture: %.1f%%\n",  node.getResponseBuffer(0)/10.0);
    Serial.printf("Temp:     %.1fC\n",   node.getResponseBuffer(1)/10.0);
    Serial.printf("EC:       %d uS/cm\n",node.getResponseBuffer(2));
    Serial.printf("pH:       %.1f\n",    node.getResponseBuffer(3)/10.0);
    Serial.printf("N:        %d mg/kg\n",node.getResponseBuffer(4));
    Serial.printf("P:        %d mg/kg\n",node.getResponseBuffer(5));
    Serial.printf("K:        %d mg/kg\n",node.getResponseBuffer(6));
  }
  delay(5000);
}
```

✅ **Serial shows all 7 values**

4b. Add BME280 reads

Merge Adafruit BME280 example into above. Print air temp/humidity too.

4c. Add WiFi + HTTPS POST

Connect to home/lab WiFi. POST JSON payload to a test endpoint (use webhook.site for a free test URL):

```
HTTPClient http;
http.begin("https://webhook.site/YOUR-UNIQUE-ID");
http.addHeader("Content-Type", "application/json");
int code = http.POST(jsonPayload);
```

✅ **Data lands at webhook.site**

4d. Add deep sleep

```
esp_sleep_enable_timer_wakeup(15 * 60 * 1000000ULL); // 15 min
esp_deep_sleep_start();
```

✅ **Current drops to µA level between wakes**

STEP 5 — Power System Integration (Day 11–12)

Disconnect USB and run from solar+battery.

1. Solar panel (+) → CN3791 SOLAR+
2. Solar panel (–) → CN3791 SOLAR–
3. LiPo (+) → CN3791 BAT+
4. LiPo (–) → CN3791 BAT–
5. CN3791 OUT+ → MT3608 IN+ (boost to 12V for probe) **and** → ESP32 VIN
6. CN3791 OUT– → all GND
7. Adjust MT3608 trimpot to exactly **12.0V** before connecting probe

Measure with multimeter:

- Battery voltage: 3.7–4.2V depending on charge
- Probe supply: 12.0V
- ESP32 VIN: comes from battery via CN3791, ~3.7–4.2V

Run device for 24 hours unattended:

- Check data still arrives at webhook.site every 15 min
- Check battery voltage in morning — should be ≥3.8V

✅ **Pass → Power system is sound**

STEP 6 — PCB Design (Day 13–20)

Once breadboard is stable, shrink to a PCB.

1. Install **KiCad 8** (free)
2. Schematic: ESP32-C3-MINI-1 module + MAX3485 + CN3791 + MT3608 equivalent + USB-C + JST-PH battery connector + BME280 on-board
3. Layout: target **50×60 mm 2-layer PCB**
4. Export Gerbers → upload to **JLCPCB**
5. Order 10 boards with JLC SMT assembly for all parts except the ESP32 module and connectors
6. Cost: **~₹4000 for 10 assembled boards** including shipping to India

Timeline: PCBs arrive ~8–10 days after order.

> You can skip Step 6 entirely for v1 prototype — just build 2-3 breadboard units for field testing. PCB is for v2.

STEP 7 — Enclosure Fabrication (Day 13–17, parallel with PCB)

1. **Measure** your assembled breadboard stack: width × length × height. Add 5 mm clearance each side.
2. **Model in FreeCAD / Fusion 360:**
   - Two-piece case: bottom (deep) + lid
   - Recessed cavity in lid for solar panel
   - Cable gland hole at bottom-center for probe cable
   - M3 heat-set inserts for lid screws
   - Silicone gasket groove around mating surface
3. **3D print** in PETG (UV-resistant) — use a local printing service (~₹300–500 for both pieces)
4. **Pot the solar panel** into the lid recess with clear UV-curing epoxy
5. **Fit gasket**, assemble with M3 screws
6. **Waterproof test:** submerge empty enclosure in bucket of water for 30 min → open → check for moisture inside

STEP 8 — Final Assembly (Day 21–23)

1. Place PCB (or breadboard stack) inside enclosure
2. Double-sided foam tape under PCB
3. Battery in its silicone-lined compartment (keep away from ESP32 heat)
4. Pass probe cable through PG7 gland → tighten gland
5. Connect cable terminals inside
6. Smoke test: close lid, power via solar, check WiFi connection
7. Seal lid with gasket + screws

STEP 9 — Field Calibration (Day 24–26)

1. **Collect 3 soil samples** from 3 different plots where device will deploy
2. **Send samples to a certified soil lab** (NABL-accredited, ~₹500/sample for N/P/K/pH)
3. **Take device readings at the exact same spots** on the same day
4. **Calculate offset/scaling factors** between device and lab values
5. **Store calibration constants in EEPROM / NVS on the device**
6. Firmware applies correction on each read

STEP 10 — Field Deployment (Day 27+)

1. Install device in target field — probe in soil, housing above ground level
2. Sun exposure must be ≥4 hours/day — avoid shade under trees
3. Ensure WiFi signal reachable (RSSI > –75 dBm)
4. Run for **14 days**
5. Monitor dashboard for:
    - Data continuity (no gaps)
    - Battery voltage trends
    - Reading sanity vs weather
6. Fix issues → reflash OTA → repeat

Critical Path Summary

Week	Milestones
Week 1	Order parts, tools, test each component individually
Week 2	Breadboard integration, firmware layers 4a–4d
Week 3	Power system + 24hr bench test; start PCB + enclosure in parallel
Week 4	PCB arrives, enclosure printed, final assembly
Week 5	Lab calibration
Week 6	Field deployment + iteration

Total: ~6 weeks to a working field prototype.

Key Design Decisions (Locked)

- **Connectivity:** WiFi-only for prototype v1 (4G variant for v2)
- **MCU:** ESP32-WROOM-32 for prototype; migrate to ESP32-C3-MINI-1 on custom PCB
- **Solar panel:** ETFE-laminated monocrystalline for 10+ year outdoor lifespan
- **Probe:** Off-the-shelf JXBS-3001 7-in-1 for v1; custom probe only if volume justifies
- **Sample interval:** 15 minutes
- **Data transmission:** HTTPS POST to KisanSaathi Vercel backend, HMAC-signed

Honest Technical Risks

- **NPK accuracy:** Dielectric-based NPK sensors measure soil EC and infer nutrient levels via proprietary formulas. Accuracy ±10–20% for trends, not lab-grade for absolute values. Market as trend monitoring, not lab replacement.
- **Battery life during monsoon:** Low solar + high humidity. Size solar/battery with 2× margin.
- **Field theft/tampering:** Consider anti-tamper design and GPS geofencing alerts.
- **Rural connectivity:** 2G/3G patchy. LoRaWAN with on-farm gateway is backup path.
- **Probe corrosion:** Stainless steel probes last 2–3 years. Design for field-replacement.

Immediate Action Items for Team

1. **Today:** Place the Robu.in order from Step 1 (soil probe has longest lead time, ~5–7 days)
2. **Confirm tool availability** — soldering iron, multimeter, 3D printing access
3. **Week 1:** Execute Step 2 (individual component validation) as parts arrive
4. **Assign roles:**

- Hardware lead: component sourcing, soldering, assembly
- Firmware lead: Arduino/PlatformIO development
- Mechanical lead: enclosure design, 3D printing
- Testing lead: calibration, field deployment

5. **Coordinate with software team** on API endpoint for data ingestion (Vercel + Supabase)

Document generated for the KisanSaathi hardware team. Source: KisanSaathi engineering planning session.
For firmware source code, PCB schematics, or enclosure STL files, contact the project lead.